



Problem Grid

Level Operators

Smooth $Ax^h = b^h$

Residual $r^h = b^h - Ax^h$

Finest grid

Level 0

Level Operators

Smooth $Ax^h = b^h$

Error correction $x^h += l^h$

Interpolation Operator

$l^h = \text{interpolate}(x^{2h})$

Coarsening Operator

Restrict $b^{2h} = \text{restrict}(r^h)$

Level Operators

Smooth $Ax^{2h} = b^{2h}$

Residual $r^{2h} = b^{2h} - Ax^{2h}$

Level 1

Level Operators

Smooth $Ax^{2h} = b^{2h}$

Error correction $x^{2h} += l^{2h}$

Interpolation Operator

$l^{2h} = \text{interpolate}(x^{4h})$

Coarsening Operator

Restrict $b^{4h} = \text{restrict}(r^{2h})$

Level 2

Coarsest grid

Coarsest level

Direct/iterative solver

Solve $Ax^{4h} = b^{4h}$